MIL-W-13241A(EL)
19 May 1965

SUPERSEDING
MIL-W-13241(SigC)
1 February 1954

MILITARY SPECIFICATION

WIRE, ELECTRICAL; ANTENNA, (WIRE WS-12( )/U and WIRE WS-13( )/U)

1. SCOPE

1.1 This specification covers bare, copper-clad steel, stranded wire having 40 percent electrical conductivity.

1.2 Classification.- The wire offered under this specification shall be as follows:

	STRANDS			
TYPE designation	NUMBER	AWG. size	NOMINAL o.d. (inch)	NOMINAL wt./1000 ft.
WS-12( )/U	7	12	0.242	128.1 lbs.
WS-13( )/U	3	12	0.174	54.8 lbs.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-W-345 Wire, Electrical, Steel, Copper-Covered.

MILITARY

MIL-C-12000 Cable, Cord, and Wire, Electric; Packaging of

FSC 6145

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STANDARDS

MILITARY

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials.- The material used shall be as specified herein. The best material commercially available for the purpose shall be used when a definite material is not designated.

3.2 Strand.- The strands shall be Type I, 0.0808 inch nominal diameter in accordance with Specification QQ-W-345.

3.3 Construction.- The wire shall be constructed of the number and kind of strands as specified in 1.2 and 3.2. The strands shall lay naturally in their true positions and not tend to unstrand at the ends when severed at any point. They shall permit restranding by hand after being forcibly unraveled (unstranded).

3.3.1 Lay.- The stranded wire shall be constructed with a left-hand lay.

3.3.2 Pitch.-

3.3.2.1 WS-12( )/U.- The pitch of the 7 strand Wire WS-12( )/U shall be from 10 to 16 times the nominal wire diameter.

3.3.2.2 WS-13( )/U.- The pitch of the 3 strand Wire WS-13( )/U shall be from 14 to 20 times the nominal wire diameter.

3.3.3 Breaking strength.-

3.3.3.1 WS-12( )/U.- The breaking strength of WS-12( )/U shall be not less than 4,946 pounds.

3.3.3.2 WS-13( )/U.- The breaking strength of WS-13( )/U shall be not less than 2,236 pounds.

3.3.4 Elongation.- The elongation of the stranded Wire WS-12( )/U and WS-13( )/U shall be not less than 1.0 per cent.

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3.4 Electrical Characteristics.-

3.4.1 D.C. resistance of strands.- The d.c. resistance when measured on samples taken not more than 25 feet from either end of any length of strand shall correspond to a resistivity not greater than 28.6 ohms per circular mil-foot at 68°F. The resistance when measured on samples taken from the mid-portion of any coil or reel shall correspond to a resistivity not greater than 26.5 ohms per circular mil-foot at 68°F. When temperatures differ from 68°F., corrections shall be based on a temperature coefficient of resistance of 0.0021 per degree Fahrenheit.

3.4 Workmanship.- Strands and completed wire shall be manufactured and assembled in a thoroughly workmanlike manner.

4. Quality Assurance Provisions

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection.- Inspection shall be classified as follows:

(a) Quality conformance inspection before preparation for delivery.(See 4.3)

(b) Quality conformance inspection of preparation for delivery. (See 4.7)

4.3 Quality conformance inspection before preparation for delivery.- The supplier shall perform the inspection specified in 4.3.1 and 4.3.2.

4.3.1 Group A inspection.- This inspection, including sampling, shall consist of the examinations of paragraph 4.4 to determine compliance with the requirements of 3.1 through 3.3.2.2, as applicable, and 3.5. Sampling shall conform to the ordinary sampling procedures of Standard MIL-STD-105, AQL-1.0% Major defects and 4.0% Minor defects.

4.3.2 Group B inspection.- This inspection, including sampling, shall consist of the inspections required by paragraphs 4.5 and 4.6 to determine compliance with paragraphs 3.3.3 , 3.3.4, & 3.4.1 respectively. Sampling shall conform to MIL-STD-105 special level S-4 for normal, tightened and reduced inspection; AQL shall be 4.0% for Major defects.

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4.4 Visual and dimensional inspection.- Strands and completed wire shall be inspected to verify that the materials and all workmanship comply with this specification.

4.4.1 Stranding.- Specimens shall be inspected for compliance with the requirements of 3.3.

4.4.2 Pitch and lay.- Specimens shall be inspected for compliance with the pitch and lay requirements of 3.3.1 through 3.3.2.2.

4.5 Breaking strength and elongation test.- Specimens shall be tested for compliance with the requirements of 3.3.3 and 3.3.4 in a tensile testing machine equipped with suitable jaws or sockets. No specimen shall have less than 10 inches of its length between the jaws or sockets. Elongation measurement shall be made on the samples at the same time that the tensile strength test is being performed. In making the elongation measurements, the length, when under a load equal to 10 per cent of the required minimum breaking strength shall be denoted as the zero length of the specimens. The final length shall be the reading when fracture occurs. The finished wire shall be considered to have fractured when one strand breaks. Any test in which the tensile strength or elongation is below the required value and in which fracture occurs within one (1) inch from the gripping jaws or sockets shall be disregarded and another adjacent specimen from the same sample shall be tested as required above.

4.6 D.C. resistance.- Specimens shall be tested for compliance with the d.c. resistance requirements of 3.4.1.

4.7 Quality conformance inspection of preparation for delivery.- Preparation for delivery shall be inspected to determine conformance to the requirements of Section 5.

5. PREPARATION FOR DELIVERY

5.1 Wires shall be preserved, packaged, packed and marked in accordance with Specification MIL-C-12000. (See 6.2).

5.2 Preservation, packaging and packing shall be level A, B, or C as specified. (See 6.2).

6. NOTES

6.1 Intended use.- The wires covered by this specification are intended for use in radio antennas.

6.2 Ordering data.- Procurement documents should specify the following:

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(a) Title, number and date of specification.
(b) Type wire and length required. (See 1.2.)
(c) Level of packaging and level of packing required (see 5.1, 5.2).
(d) Place of final inspection.

6.3 Nomenclature.- The parenthesis in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: WB-12W/U. As soon as possible after the award of contract, the contractor should apply to the Government office specified in the contract for such information.

6.4 Definitions

6.4.1 Inspection lot.- An inspection lot shall consist of all units of product, of one type, produced under essentially the same conditions, and submitted for acceptance at the same time.

6.4.2 Unit of product.- A unit of product is the continuous length of wire contained on a single reel, a spool, or in a package.

6.4.3 Specimen.- A specimen consists of the individual piece of the unit of product upon which the test is made.

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Project Nr. 6145-A039

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES" IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity)	DATE

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